

I. INTRODUCTION

II. SYSTEM MODEL

As presented in Fig. ??, we investigate a cloud-edge-device collaborative system in LAWNs, consisting of one cloud server, G mobile edge servers, and U UAV devices. Due to the difference in computing resources among the cloud server, edge servers, and UAV devices, large AI models with different parameter scales are deployed respectively on the three ends. The cloud server collects data to fine-tune the cloud-side model, and then distributes the newly learned knowledge to edge servers and UAVs. Due to privacy concerns, raw data cannot be directly transmitted to edge servers or UAVs. To address this, we propose a LoRA-based parameter-efficient updating model, which operates as follows:

- *Cloud model fine-tuning*: The cloud server uses the LoRA technique to fine-tune the cloud-side large model using the collected data, obtaining the updated cloud LoRA parameters, denoted as θ^C .
- *Cloud-to-edge transmission*: The cloud server transmits the updated LoRA parameters θ^C to each edge server.
- *Edge LoRA transferring*: Upon receiving the LoRA parameters θ^C from the cloud server, each edge server resizes it to fit the edge-side large AI model using the proposed LoRA transformation method (in Section III.B), obtaining the updated edge LoRAs, denoted as θ^E .
- *Edge-to-UAV transmission*: Each edge server transmits its updated LoRA parameters to the associated UAVs.
- *UAV LoRA transferring*: After receiving the LoRA parameters, each UAV resizes it to fit the device-side local large AI model using the proposed LoRA transformation method (in Section III.C), obtaining the updated UAV LoRAs, denoted as θ^U .

Through these five steps, the knowledge in the cloud data is transferred to the edge servers and UAVs while protecting the data privacy.

The set of edge servers and UAVs are denoted as $\mathcal{G} = \{1, \dots, G\}$ and $\mathcal{U} = \{1, \dots, U\}$, respectively. The mission of the LoRA parameter update should be completed within time period T , which is divided into N time interval, each with a duration of $\tau = \frac{T}{N}$, where the set of time interval is denoted as $\mathcal{N} = \{1, \dots, N\}$. The cloud server is located at a fixed position $\mathbf{C} = \{c_x, c_y, H_c\}$. We assume that the edge servers (e.g., ground vehicles) navigate on the internal roads within the mission region, and the corresponding UAVs navigate on the pre-designed low-altitude corridors. We assume that the navigation information, including the positions and velocities of the edge serves and UAVs are known. The positions of the g -th edge server and the u -th UAV at time interval n are denoted as $\mathbf{E}_g[n] = \{e_g[n], 0\}$ and $\mathbf{U}_u[n] = \{\mathbf{u}_u[n], H\}$, where $e_g[n] = \{e_g^x[n], e_g^y[n]\}$ is the two-dimensional (2D) coordinates of the g -th edge server, $\mathbf{u}_u[n] = \{u_u^x[n], u_u^y[n]\}$ represents the 2D coordinates of the u -th UAV, and H is the flying altitude of the UAVs.

A. Cloud Update and Transmission

The cloud server uses the newly collected data to update the LoRA parameter θ^C . The cloud model updates are performed

continuously. At the beginning of each interval, the current LoRA parameters θ^C is transmitted to the edge servers. Specifically, the cloud server updates θ^C during the previous time interval, then multicasts the updated parameters θ^C to the edge servers in the current time interval. Thus, we ignore the time for the cloud server to update the cloud LoRA parameters in this work. The updated LoRA parameters are formatted into data packets, with a size of d_C bits, and transmitted as the signal s_C over wireless channels. We consider that the cloud server is equipped with M -antenna linear-array. The transmitted signal by the cloud server at time interval n can be given by

$$\mathbf{x}_C[n] = \mathbf{w}_C[n] s_C[n], \quad (1)$$

where $\mathbf{w}_C[n]$ denotes the transmit beamforming vector which is limited by a maximum transmit power $P_{\max}^C$ as $\|\mathbf{w}_C[n]\|^2 \leq P_{\max}^C, \forall n$, and $s_C[n]$ is the data symbol with $\mathbb{E}\{|s_C[n]|^2\} = 1$.

The channel between the cloud server and the g -th edge server $\mathbf{h}_g[n]$ is modeled as Rician fading, which is given by

$$\mathbf{h}_g[n] = g_g[n] \left(\sqrt{\frac{k_g}{k_g + 1}} \mathbf{a}_g^{\text{LoS}}[n] + \sqrt{\frac{1}{k_g + 1}} \mathbf{a}_g^{\text{NLoS}}[n] \right), \quad (2)$$

where $g_g[n] = \frac{g_0}{\|\mathbf{C} - \mathbf{E}_g[n]\|}$ is the path-loss, g_0 denotes the channel power gain at a reference distance, k_g is the Rician factor of the channel between the cloud server and the g -th edge server, and $\mathbf{a}_g^{\text{LoS}}[n]$ is the steering vector, which is given by

$$\mathbf{a}_g^{\text{LoS}}[n] = [1, e^{-j\phi_g[n]}, \dots, e^{-j(M-1)\phi_g[n]}]^T, \quad (3)$$

where $\phi_g[n] = \frac{2\pi d}{\lambda} \cos \theta(\mathbf{C}, \mathbf{G}_g[n])$. Here, d and λ represent the spacing between two adjacent antennas and the carrier wavelength, respectively, and $\theta(\mathbf{C}, \mathbf{G}_g[n])$ denotes the AoD from the cloud server to the g -th edge server, which can be expressed as

$$\theta(\mathbf{C}, \mathbf{G}_g[n]) = \arccos \frac{H_c}{\|\mathbf{C} - \mathbf{G}_g[n]\|}. \quad (4)$$

Besides, $\mathbf{a}_g^{\text{NLoS}}[n]$ denotes the NLoS component of the channel between the cloud server and the g -th edge server, which has a zero mean and unit variance. Then, the received signal of the g -th edge server at time interval n can be given by

$$y_g[n] = \mathbf{h}_g^H[n] \mathbf{x}_{\text{Cloud}}[n] + n_g[n], \quad (5)$$

where $n_g[n] \sim \mathcal{CN}(0, \sigma_g^2)$ is the additive white gaussian noise (AWGN) noise. The achievable data rate can be given by

$$R[n] = B \log_2 \left(1 + \min_{g \in \mathcal{G}} \frac{|\mathbf{h}_g^H[n] \mathbf{w}_C[n]|^2}{\sigma_g^2} \right), \quad (6)$$

where B is the system bandwidth. Accordingly, the latency for transmitting the cloud LoRA parameters to the g -th edge server can be given by

$$L_C^{\text{tran}}[n] = \frac{d_C[n]}{R[n]}. \quad (7)$$

Then, the energy consumption for cloud LoRA transmission can be expressed as

$$E_C[n] = \|\mathbf{w}_C[n]\|^2 L_C^{\text{tran}}[n]. \quad (8)$$

B. Edge Update and Transmission

Upon receiving the updated LoRA θ^C from the cloud server, each edge server transforms the LoRA θ^C to their respective LoRA θ^E . In this work, we use a CNN to map the LoRA θ^C to the LoRA θ^E . The CNN Upon receiving the updated LoRA θ^C from the cloud server, each edge server transforms the LoRA θ^C to their respective LoRA θ^E . In this work, we use a CNN to map the LoRA θ^C to the LoRA θ^E . Specifically, the CNN is designed as a lightweight one-dimensional row-wise mapper. For each LoRA matrix, each row vector is treated as a one-dimensional feature sequence and is fed into a three-layer 1D convolutional network. The convolutional branch consists of a $1 \rightarrow H$ convolution, an $H \rightarrow H$ convolution, and an $H \rightarrow 1$ convolution, where H denotes the number of hidden channels. Each of the first two convolutional layers is followed by a GELU activation function. Since the cloud model and the edge model may have different hidden dimensions, the output of the convolutional branch is resized to the target LoRA dimension through linear interpolation. In addition, a linear shortcut projection is introduced to preserve the global mapping between the source and target LoRA spaces. Therefore, for a source LoRA row vector $\mathbf{x}^C \in \mathbb{R}^{d_C}$, the transferred edge LoRA row vector $\mathbf{x}^E \in \mathbb{R}^{d_E}$ is obtained as

$$\mathbf{x}^E = \sigma(\lambda) \mathbf{W}_s \mathbf{x}^C + (1 - \sigma(\lambda)) \text{Interp}(\text{Conv}_3(\phi(\text{Conv}_2(\phi(\text{Conv}_1(\mathbf{x}^C)))))), \quad (9)$$

where $\phi(\cdot)$ is the GELU activation function, $\text{Interp}(\cdot)$ denotes linear interpolation to the target dimension d_E , $\mathbf{W}_s$ is the shortcut projection matrix, λ is a learnable mixing parameter, and $\sigma(\cdot)$ is the sigmoid function. In our implementation, we set $H = 16$ and use a convolution kernel size of 5. Moreover, separate CNN mappers are trained for different LoRA modules and for the two LoRA factors, i.e., the A and B matrices.xxxxxx.

The computational workload for transferring at the edge server is denoted as $F_{\text{tf},E}[n]$. Meanwhile, we assume that the g -th edge server has a total of K_g GPU cores, with each core having a computing capability of f_{Edge} . Then, the latency for LoRA transferring at the g -th edge server can be given by

$$L_g^{\text{upd}}[n] = \frac{F_{\text{fer},\text{Edge}}[n] \mu_g}{\alpha_g[n] f_{\text{Edge}}} + \xi_g, \quad (10)$$

where $\alpha_g[n]$ represents the scheduling number of the GPU cores of the g -th edge server during the time interval n , μ_g and ξ_g are constant coefficients that characterize the computational performance of the GPU on the g -th edge server. Specifically, μ_g denotes the computational efficiency coefficient, representing the average number of GPU clock cycles required per unit computational workload, ξ_g represents the kernel launch latency, context switching delay, memory allocation time, and device synchronization overhead.

Then, the g -th edge server transmits the LoRA θ^E to the associated UAVs. Let $\beta_{g,u}[n]$ denote the association between the g -th edge server and the u -th UAV, where $\beta_{g,u}[n] = 1$ represents the u -th UAV is associated with the g -th edge server and receives the LoRA θ^E from the associated edge server, and otherwise $\beta_{g,u}[n] = 0$. Due to the capacity constraints, each edge server can only serve $\beta_{\max}$ UAVs during each interval. Hence, we have $\sum_{u \in \mathcal{U}} \beta_{g,u}[n] \leq \beta_{\max}, \forall g$. To ensure fairness in transmission, we assume that each UAV can be associated with only one edge server, i.e., $\sum_{g \in \mathcal{G}} \beta_{g,u}[n] = 1$.

We assume that the edge servers and UAVs are equipped with single antenna, and the frequency division multiple access protocol is adopted. Specifically, the bandwidth is allocated to each edge server, and $\gamma_g[n]$ represents the allocation ratio of the bandwidth for the g -th edge server, where $0 < \gamma_g[n] < 1$.

The updated LoRA parameters at the g -th edge server are also formatted into data packets, with a size of d_E bits, and can be transmitted as the data symbol $s_g[n]$ over wireless channels, with $\mathbb{E}\{|s_g[n]|^2\} = 1$. Then, the transmit signal by the g -th edge server is given by $x_g[n] = \sqrt{p_g[n]} s_g[n]$, where $p_g[n] \leq P_{\max}^E$ is the transmit power of the g -th edge server with a maximum transmit power $P_{\max}^E$. Accordingly, the received signal at the u -th UAV by the g -th edge server can be expressed as

$$y_{u,g}[n] = \beta_{g,u}[n] (h_{g,u}[n] x_g[n] + n_u[n]), \quad (11)$$

where $n_u[n] \sim \mathcal{CN}(0, \sigma_u^2)$ denotes the AWGN at the u -th UAV, and $h_{g,u}[n]$ is the channel gain between the g -th edge server and the u -th UAV, which is modeled as Rician fading

$$h_{g,u}[n] = g_{g,u}[n] \left(\sqrt{\frac{k_{g,u}}{k_{g,u} + 1}} + \sqrt{\frac{1}{k_{g,u} + 1}} a_{g,u}^{\text{NLoS}}[n] \right), \quad (12)$$

where $g_{g,u}[n] = \frac{g_0}{\| \mathbf{E}_g[n] - \mathbf{U}_u[n] \|^2}$ is the path-loss, $k_{g,u}$ is the Rician factor of the channel between the g -th edge server and the u -th UAV, and $a_{g,u}^{\text{NLoS}}[n] \sim \mathcal{CN}(0, 1)$ is the NLoS component.

Hence, the achievable data rate from the g -th edge server to the u -th UAV can be given by

$$R_{g,u}[n] = \gamma_g[n] B \log_2 \left(1 + \frac{\beta_{g,u}[n] p_g[n] |h_{g,u}[n]|^2}{\sigma_u^2} \right). \quad (13)$$

As a result, the latency for transmitting the edge LoRA to the u -th UAV can be given by

$$L_u^{\text{tran}}[n] = \frac{d_{\text{Edge}}[n]}{\sum_{g=1}^G R_{g,u}[n]}. \quad (14)$$

Then, the total energy consumption of the g -th edge server is given by

$$E_g[n] = \alpha_g[n] p_{\text{Edge}} L_g^{\text{upd}}[n] + p_g[n] \max_{\mathcal{U}_g[n]} L_u^{\text{tran}}[n], \quad (15)$$

where p_{Edge} denotes the GPU computing power of edge servers, and $\mathcal{U}_g[n] = \{u | \beta_{g,u}[n] = 1\}$.

C. UAV Update Model

After receiving the updated LoRA from the associated edge server, the UAVs transform it to fit the local large AI model using a similar CNN model. The computational workload for transferring the E-LoRA transmitted by the g -th edge server is $F_{\text{fer},g}[n]$. Similarly, the u -th UAV has a total of K_u GPU cores, with each core having the computing capability of f_{UAV} . Hence, the latency for LoRA transferring of the u -th UAV is expressed as

$$L_u^{\text{upd}}[n] = \frac{F_{\text{fer},g^u}[n]\mu_u}{\alpha_u[n]f_{\text{UAV}}} + \xi_u, \quad (16)$$

where $g^u = \{g | \beta_{g,u} = 1\}$, $\alpha_u[n]$ represents the GPU computational capability of the u -th UAV, and μ_u and ξ_u are coefficients that characterize the computational performance of the GPU on the u -th UAV. Hence, the total energy consumption of the u -th UAV can be given by

$$E_u[n] = \alpha_u[n]p_{\text{UAV}}L_u^{\text{upd}}[n], \quad (17)$$

where p_{UAV} denotes the GPU computing power of UAVs.

D. Problem Formulation

To facilitate the distributed update and deployment of large AI models for LAE applications, we aim to minimize the time-averaged energy consumption during the update process

$$(\mathbf{P1}) : \min_{\Psi} \frac{1}{N} \sum_{n=1}^N \left(E_{\text{CCS}}[n] + \sum_{g=1}^G E_g[n] + \sum_{u=1}^U E_u[n] \right) \quad (18a)$$

$$\text{s.t. } \|\mathbf{w}_{\text{CCS}}[n]\|^2 \leq P_{\text{max}}^{\text{CCS}}, \quad \forall n \in \mathcal{N}, \quad (18b)$$

$$0 \leq \alpha_g[n] \leq K_g, \quad \forall \alpha_g[n] \in \mathbb{I}, \quad \forall n \in \mathcal{N}, \quad (18c)$$

$$\beta_{g,u}[n] \in \{0, 1\}, \quad \forall g \in \mathcal{G}, \forall u \in \mathcal{U}, \forall n \in \mathcal{N}, \quad (18d)$$

$$\sum_{u \in \mathcal{U}} \beta_{g,u}[n] \leq \beta_{\text{max}}, \quad \forall g \in \mathcal{G}, \forall n \in \mathcal{N}, \quad (18e)$$

$$\sum_{g \in \mathcal{G}} \beta_{g,u}[n] = 1, \quad \forall u \in \mathcal{U}, \forall n \in \mathcal{N}, \quad (18f)$$

$$0 < \gamma_g[n] < 1, \quad \forall g \in \mathcal{G}, \forall n \in \mathcal{N}, \quad (18g)$$

$$\sum_{g=1}^G \gamma_g[n] \leq 1, \quad \forall n \in \mathcal{N}, \quad (18h)$$

$$p_g[n] \leq P_{\text{max}}^{\text{UGV}}, \quad \forall g \in \mathcal{G}, \forall n \in \mathcal{N}, \quad (18i)$$

$$0 \leq \alpha_u[n] \leq K_u, \quad \forall \alpha_u[n] \in \mathbb{I}, \quad \forall n \in \mathcal{N}, \quad (18j)$$

$$L_g[n] \leq \tau, \quad \forall g \in \mathcal{G}, \forall n \in \mathcal{N}, \quad (18k)$$

$$L_{\text{CCS},g}^{\text{tran}}[n] \leq t_{\text{th}}, \quad \forall g \in \mathcal{G}, \forall n \in \mathcal{N}, \quad (18l)$$

where $L_g[n]$ denotes the total update latency associated with the g -th UGV, which ensures that the update latency cannot exceed the duration of each interval, and is given by

$$L_g[n] = L_{\text{CCS}}^{\text{upd}}[n] + L_{\text{CCS},g}^{\text{tran}}[n] + L_g^{\text{upd}}[n] + \max_{u \in \mathcal{U}_g[n]} (L_u^{\text{tran}}[n] + L_u^{\text{upd}}[n]), \quad (19)$$

and t_{th} represents the latency threshold for the G-LoRA deployment of the CCS-to-UGV links.

III. PROBLEM TRANSFORMATION AND SOLUTION

A. Problem Transformation

We can observe that, there are no cross-interval dependencies or shared variables in problem **P1**. Hence, it can be safely decomposed into N independent single-interval subproblems, each solved locally and in parallel. Let subproblem **P1.n** denote the subproblem related to the n -th interval, which is then given by

$$(\mathbf{P1.n}) : \min_{\Psi} E_{\text{CCS}}[n] + \sum_{g=1}^G E_g[n] + \sum_{u=1}^U E_u[n] \quad (20a)$$

$$\text{s.t. } \|\mathbf{w}_{\text{CCS}}[n]\|^2 \leq P_{\text{max}}^{\text{CCS}}, \quad (20b)$$

$$0 \leq \alpha_g[n] \leq K_g, \quad \forall \alpha_g[n] \in \mathbb{I}, \quad (20c)$$

$$\beta_{g,u}[n] \in \{0, 1\}, \quad \forall g \in \mathcal{G}, \forall u \in \mathcal{U}, \quad (20d)$$

$$\sum_{u \in \mathcal{U}} \beta_{g,u}[n] \leq \beta_{\text{max}}, \quad \forall g \in \mathcal{G}, \quad (20e)$$

$$\sum_{g \in \mathcal{G}} \beta_{g,u}[n] = 1, \quad \forall u \in \mathcal{U}, \quad (20f)$$

$$0 < \gamma_g[n] < 1, \quad \forall g \in \mathcal{G}, \quad (20g)$$

$$\sum_{g=1}^G \gamma_g[n] \leq 1, \quad (20h)$$

$$p_g[n] \leq P_{\text{max}}^{\text{UGV}}, \quad \forall g \in \mathcal{G}, \quad (20i)$$

$$0 \leq \alpha_u[n] \leq K_u, \quad \forall \alpha_u[n] \in \mathbb{I}, \quad (20j)$$

$$L_g[n] \leq \tau, \quad \forall g \in \mathcal{G}, \quad (20k)$$

$$L_{\text{CCS},g}^{\text{tran}}[n] \leq t_{\text{th}}, \quad \forall g \in \mathcal{G}. \quad (20l)$$

B. Subproblem S2: CCS Beamforming

Given all other variables, we optimize the transmit beamforming of the CCS $\mathbf{w}$. Hence, we can restate the problem **P1.n** as

$$(\mathbf{S2.n}) : \min_{\Psi} \|\mathbf{w}_{\text{CCS}}[n]\|^2 \max_{g \in \mathcal{G}} L_{\text{CCS},g}^{\text{tran}}[n] + \Psi_{\text{S2}}[n] \quad (21a)$$

$$\text{s.t. } \|\mathbf{w}_{\text{CCS}}[n]\|^2 \leq P_{\text{max}}^{\text{CCS}}, \quad (21b)$$

$$L_g[n] \leq \tau, \quad \forall n \in \mathcal{N}, \quad (21c)$$

$$L_{\text{CCS},g}^{\text{tran}}[n] \leq t_{\text{th}}, \quad \forall g \in \mathcal{G}, \quad (21d)$$

where $\Psi_{\text{S2}}[n]$ is a constant value related to other fixed variables, and is expressed as

$$\Psi_{\text{S2}}[n] = \alpha_{\text{CCS}}[n]p_{\text{CCS}}L_{\text{CCS}}^{\text{upd}}[n] + \sum_{g=1}^G E_g[n]. \quad (22)$$

Problem **S2.n** cannot be solved due to the non-convex objective and the coupling of the beamforming vector and the worst-case CCS transmission latency. Hence, we first introduce an auxiliary variable $\chi[n]$, which satisfied

$$\|\mathbf{w}_{\text{CCS}}[n]\|^2 L_{\text{CCS},g}^{\text{tran}}[n] \leq \chi[n], \quad \forall g \in \mathcal{G}. \quad (23)$$

Meanwhile, the transmission threshold t_{th} is less than the duration of time interval τ . Thus, constraint (20j) is equivalent to constraint (20k). Then, we can rewritten problem **S2.n** as

$$(\mathbf{S2-1.n}) : \min_{\mathbf{w}} \chi[n]$$

$$\text{s.t. } \|\mathbf{w}_{\text{CCS}}[n]\|^2 \leq P_{\max}^{\text{CCS}}, \quad (24a)$$

$$L_{\text{CCS},g}^{\text{tran}}[n] \leq t_{\text{th}}, \quad \forall g \in \mathcal{G}, \quad (24b)$$

$$\|\mathbf{w}_{\text{CCS}}[n]\|^2 L_{\text{CCS},g}^{\text{tran}}[n] \leq \chi[n], \quad \forall g \in \mathcal{G}. \quad (24c)$$

To handle the non-convexity constraints in **S2-1.n**, we first rewrite constraint (20k) as

$$\log_2 \left(1 + \frac{|\mathbf{h}_g^H[n] \mathbf{w}_{\text{CCS}}[n]|^2}{\sigma_g^2} \right) \geq \frac{d_{\text{CCS}}[n]}{B t_{\text{th}}}, \quad (25)$$

which is equivalent to

$$|\mathbf{h}_g^H[n] \mathbf{w}_{\text{CCS}}[n]|^2 \geq \Psi_{\text{S2-2}}[n], \quad (26)$$

where

$$\Psi_{\text{S2-2}}[n] = \sigma_g^2 \left(2^{\frac{d_{\text{CCS}}[n]}{B t_{\text{th}}}} - 1 \right). \quad (27)$$

Let $\mathbf{W}[n] = \mathbf{w}_{\text{CCS}}[n] \mathbf{w}_{\text{CCS}}^H[n]$, where $\text{tr}(\mathbf{W}[n]) \leq P_{\max}^{\text{CCS}}$ and $\text{rank}(\mathbf{W}[n]) = 1$. Through semi-definite relaxation, constraint (26) can be transformed as

$$\text{tr}(\mathbf{h}_g[n] \mathbf{h}_g^H[n] \mathbf{W}[n]) \geq \Psi_{\text{S2-2}}[n], \quad (28)$$

$$\mathbf{W}[n] \succeq \mathbf{0}. \quad (29)$$

Then, we rewrite constraint (34d) as

$$R_g[n] \geq \frac{\|\mathbf{w}_{\text{CCS}}[n]\|^2 \cdot d_{\text{CCS}}[n]}{\chi[n]}, \quad \forall g \in \mathcal{G}. \quad (30)$$

Note that $R_g[n]$ is a concave function of $\mathbf{w}_{\text{CCS}}[n]$, but the right-hand side of (30) is convex in $\mathbf{w}_{\text{CCS}}[n]$. Thus, the constraint is non-convex. At iteration l of successive convex approximation (SCA), given the current beamforming vector $\mathbf{w}_{\text{CCS}}^{(l)}[n]$, we construct a convex lower bound for $R_g[n]$ using its first-order Taylor expansion. Specifically, let $s_g(\mathbf{w}_{\text{CCS}}[n]) = |\mathbf{h}_g^H[n] \mathbf{w}_{\text{CCS}}[n]|^2$. Then, the gradient of $R_g[n]$ with respect to $\mathbf{w}_{\text{CCS}}[n]$ at $\mathbf{w}_{\text{CCS}}^{(l)}[n]$ is expressed as

$$\nabla_{\mathbf{w}} R_g[n] \big|_{\mathbf{w}_{\text{CCS}}^{(l)}[n]} = \frac{B}{\ln 2} \cdot \frac{2 \mathbf{h}_g[n] \mathbf{h}_g^H[n] \mathbf{w}_{\text{CCS}}^{(l)}[n]}{s_g(\mathbf{w}_{\text{CCS}}^{(l)}[n]) + \sigma_g^2} \quad (31)$$

Thus, the lower bound of $R_g[n]$ can be given by

$$\begin{aligned} R_g[n] &\geq R_g^{(l)}[n] \\ &+ \Re \left\{ \left(\nabla_{\mathbf{w}} R_g[n] \big|_{\mathbf{w}_{\text{CCS}}^{(l)}[n]} \right)^H (\mathbf{w}_{\text{CCS}}[n] - \mathbf{w}_{\text{CCS}}^{(l)}[n]) \right\} \\ &\triangleq \tilde{R}_g^{(l)}[n]. \end{aligned} \quad (32)$$

Substituting (32) into (30), we obtain the following rearranged convex constraint

$$\|\mathbf{w}_{\text{CCS}}[n]\|^2 \cdot d_{\text{CCS}}[n] \leq \chi[n] \cdot \tilde{R}_g^{(l)}[n]. \quad (33)$$

Accordingly, problem **S2-1.n** is now transformed into the following problem

$$\begin{aligned} (\mathbf{S2-2.n}) : \quad &\min_{\mathbf{w}} \quad \chi[n] \\ \text{s.t.} \quad &\text{tr}(\mathbf{W}[n]) \leq P_{\max}^{\text{CCS}}, \end{aligned} \quad (34a)$$

$$\text{tr}(\mathbf{h}_g[n] \mathbf{h}_g^H[n] \mathbf{W}[n]) \geq \Psi_{\text{S2-2}}[n], \quad \forall g, \quad (34b)$$

$$\mathbf{W}[n] \succeq \mathbf{0}, \quad (34c)$$

$$\|\mathbf{w}_{\text{CCS}}[n]\|^2 \cdot d_{\text{CCS}}[n] \leq \chi[n] \cdot \tilde{R}_g^{(l)}[n], \quad \forall g. \quad (34d)$$

Problem **S2-2.n** is convex, which can be effectively solved.

C. Subproblem S3: UGV GPU Scheduling

Similar to subproblem S1, when fixed other variables, we optimize the GPU scheduling of UGVs. The problem is restated as

$$\begin{aligned} (\mathbf{S3.n}) : \quad &\min_{\Psi} \quad \sum_{g=1}^G (\alpha_g[n] p_{\text{UGV}} L_g^{\text{upd}}[n]) + \Psi_{\text{S3}}[n] \\ \text{s.t.} \quad &(20b) \text{ and } (20j), \end{aligned}$$

where

$$\Psi_{\text{S3}}[n] = E_{\text{CCS}}[n] + \sum_{g=1}^G \left(p_g[n] \max_{\mathcal{U}_g[n]} L_u^{\text{tran}}[n] \right). \quad (36)$$

Similarly, we relax the integer constraint (20b) as

$$0 \leq \alpha_g[n] \leq K_g, \quad \forall g. \quad (37)$$

Substituting into problem **S3.n**, we can obtain the following convex problem **S3-1.n** as

$$\begin{aligned} (\mathbf{S3-1.n}) : \quad &\min_{\Psi} \quad \sum_{g=1}^G (\alpha_g[n] p_{\text{UGV}} L_g^{\text{upd}}[n]) + \Psi_{\text{S3}}[n] \\ \text{s.t.} \quad &(20j) \text{ and } (37). \end{aligned}$$

Problem **S3-1.n** can be solved efficiently by conventional solvers. Moreover, the close-form solution can be derived by enforcing the delay constraint as

$$\alpha_g^*[n] = \max \left(0, \frac{F_{\text{fer,UGV}}[n] \mu_{\text{UGV}}}{f_{\text{UGV}} (\tau - \xi_{\text{UGV}})} \right). \quad (39)$$

D. Subproblem S4: UGV-UAV Association

When fixing other variables, we optimize the UGV-UAV association and reformulate the following subproblem

$$\begin{aligned} (\mathbf{S4.n}) : \quad &\min_{\Psi} \quad \sum_{g=1}^G \left(p_g[n] \max_{\mathcal{U}_g[n]} L_u^{\text{tran}}[n] \right) + \Psi_{\text{S}}[n] \\ \text{s.t.} \quad &(20c), (20d), (20e), \text{ and } (20j), \end{aligned}$$

where

$$\Psi_{\text{S}}[n] = E_{\text{CCS}}[n] + \sum_{g=1}^G (\alpha_g[n] p_{\text{UGV}} L_g^{\text{upd}}[n]). \quad (41)$$

Since $p_g[n]$ and $\Psi_{\text{S}}[n]$ are fixed constant, **S4.n** is equivalent to

$$\begin{aligned} (\mathbf{S4.n}) : \quad &\max_{\Psi} \min_{\mathbf{w}} \quad \sum_{g=1}^G R_g[n] \\ \text{s.t.} \quad &(20c), (20d), (20e), \text{ and } (20j), \end{aligned}$$

E. Subproblem S5: UGV Bandwidth Allocation

Fixing others, we optimize the bandwidth allocation of UGVs. Then, the problem can be simplified as

$$(\mathbf{S5}.n) : \min_{\Psi} \sum_{g=1}^G \left(p_g[n] \max_{\mathcal{U}_g[n]} L_u^{\text{tran}}[n] \right) + \Psi_S[n] \\ \text{s.t.} \quad (20f) \text{ and } (20j),$$

Problem **S5.n** is convex, and can be solved by conventional solver.

F. Subproblem S6: UGV Transmit Power

$$(\mathbf{S6}.n) : \min_{\Psi} \sum_{g=1}^G \left(p_g[n] \max_{\mathcal{U}_g[n]} L_u^{\text{tran}}[n] \right) + \Psi_S[n] \\ \text{s.t.} \quad (20h) \text{ and } (20j),$$

Since the objective function and constraints are non-convex and non-linear in $p_g[n]$, we employ the SCA to iteratively solve the problem. Firstly, to handle the max operator, we introduce auxiliary variables $\zeta_g[n]$ satisfies

$$t_g[n] \geq L_u^{\text{tran}}[n], \quad \forall u \in U_g[n] \quad (45)$$

Then, the problem is reformulated as

$$(\mathbf{S6-1}.n) : \min_{p_g[n], \zeta_g[n]} \sum_{g=1}^G p_g[n] \cdot \zeta_g[n] \quad (46)$$

subject to:

$$t_g[n] \geq \frac{d_{\text{UGV}}[n]}{\sum_{g'=1}^G R_{g',u}[n]}, \quad \forall u \in U_g[n] \quad (47)$$

$$0 \leq p_g[n] \leq P_{\text{UGV}}^{\text{max}} \quad (48)$$

$$L_{\text{max}}[n] \leq \tau \quad (49)$$

Define $a_{g',u} = \frac{\beta_{g',u}[n] |h_{g',u}[n]|^2}{\sigma_u^2}$. Then,

$$R_{g',u}[n] = \gamma_{g'}[n] B \log_2(1 + a_{g',u} p_{g'}[n]) \quad (50)$$

This function is concave in $p_{g'}[n]$. At the current iterate $p_{g'}^{(l)}[n]$, its first-order Taylor expansion is:

$$\tilde{R}_{g',u}^{(l)}[n] = R_{g',u}^{(l)}[n] + \nabla_{p_{g'}} R_{g',u}[n] \big|_{p_{g'}^{(l)}} \cdot (p_{g'}[n] - p_{g'}^{(l)}[n]) \quad (51)$$

where

$$\nabla_{p_{g'}} R_{g',u}[n] = \gamma_{g'}[n] B \cdot \frac{1}{\ln 2} \cdot \frac{a_{g',u}}{1 + a_{g',u} p_{g'}[n]} \quad (52)$$

Since $R_{g',u}[n]$ is concave, we have $R_{g',u}[n] \leq \tilde{R}_{g',u}^{(l)}[n]$, and thus:

$$L_u^{\text{tran}}[n] = \frac{d_{\text{UGV}}[n]}{\sum_{g'} R_{g',u}[n]} \geq \frac{d_{\text{UGV}}[n]}{\sum_{g'} \tilde{R}_{g',u}^{(l)}[n]} \quad (53)$$

Therefore, the constraint $t_g[n] \geq L_u^{\text{tran}}[n]$ can be approximated by:

$$t_g[n] \geq \frac{d_{\text{UGV}}[n]}{\sum_{g'} \tilde{R}_{g',u}^{(l)}[n]} \quad (54)$$

which is linear in $p_g[n]$. The objective function contains the bilinear term $p_g[n] \cdot t_g[n]$. To convexify it, we use the AM-GM inequality:

$$p_g[n] \cdot t_g[n] \leq \frac{1}{2} \left(\frac{p_g[n]^2}{p_g^{(l)}[n]} + \frac{t_g[n]^2}{t_g^{(l)}[n]} \right), \quad \text{for } p_g^{(l)}[n] > 0, t_g^{(l)}[n] > 0 \quad (55)$$

Thus, at iteration l , we approximate the objective as:

$$\sum_{g=1}^G \frac{1}{2} \left(\frac{p_g[n]^2}{p_g^{(l)}[n]} + \frac{t_g[n]^2}{t_g^{(l)}[n]} \right) \quad (56)$$

which is a convex quadratic function. At iteration l , given the current solution $(p^{(l)}, t^{(l)})$, we solve the following convex subproblem:

$$(\mathbf{S6-SCA}.n)^{(l)} : \min_{p_g[n], t_g[n]} \sum_{g=1}^G \frac{1}{2} \left(\frac{p_g[n]^2}{p_g^{(l)}[n]} + \frac{t_g[n]^2}{t_g^{(l)}[n]} \right) \quad (57)$$

subject to:

$$t_g[n] \geq \frac{d_{\text{UGV}}[n]}{\sum_{g'=1}^G \tilde{R}_{g',u}^{(l)}[n]}, \quad \forall u \in U_g[n] \quad (58)$$

$$0 \leq p_g[n] \leq P_{\text{UGV}}^{\text{max}} \quad (59)$$

$$L_{\text{max}}[n] \leq \tau \quad (60)$$

This is a convex optimization problem (QP or SOCP) and can be efficiently solved using standard solvers such as CVX, MOSEK, or Gurobi.